MILITARY SPECIFICATION SHEET

PLASTIC SHEET, LAMINATED, METAL-CLAD (FOR PRINTED WIRING BOARDS), BASE MATERIAL GE, GLASS BASE, WOVEN, MAJORITY DIFUNCTIONAL EPOXY RESIN, GENERAL PURPOSE, COPPER-CLAD

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the plastic sheets described herein shall consist of this specification and the latest issue of MIL-P-13949.

REQUIREMENTS:

Property values: See table I.

TABLE I. Property values.

Property to be tested	Test condition 1/	Core thickness: Less than 0.020	Core thickness: 0.020 in. and over
1. Peel strength, minimum in lbs-per-inch width			
A. 1/8-oz/ft^2 copper			
As received	A	4.5	4.5
After thermal stress	A	4.5	4.5
At elevated temperature	125°C	3.5	3.5
After exposure to processing solutions	A	4.5	4.5
B. 1/4-oz/ft^2 copper			
As received	A	5.0	5.5
After thermal stress	A	5.0	5.0
At elevated temperature	125°C	4.0	4.0
After exposure to processing solutions	A	4.5	4.5
C. 3/8-oz/ft^2 copper			
As received	A	5.5	6.0
After thermal stress	A	5.5	5.5
At elevated temperature	125°C	5.0	5.0
After exposure to processing solutions	A	5.0	5.0
D. 1/2-oz/ft^2 copper			
As received	A	4.5	6.0
After thermal stress	A	4.5	6.0
At elevated temperature	125°C	4.0	4.0
After exposure to processing solutions	A	3.0	4.5
E. 3/4-oz/ft^2 copper			
As received	A	5.0	7.0
After thermal stress	A	5.0	7.0
At elevated temperature	125°C	4.5	4.5
After exposure to processing solutions	A	4.0	5.0

See footnotes at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Core thickness: Less than 0.020	Core thickness: 0.020 in. and over
F. 1-oz/ft^2 copper			
As received	A	6.0	8.0
After thermal stress	A	6.0	8.0
At elevated temperature	125°C	5.0	5.0
After exposure to processing solutions	A	5.0	7.0
G. 2-oz/ft^2 copper			
As received	A	8.0	11.0
After thermal stress	A	8.0	11.0
At elevated temperature	125°C	6.0	6.0
After exposure to processing solutions	A	6.0	9.0
H. 3-, 4-, and 5-oz/ft^2 copper			
As received	A	9.0	12.0
After thermal stress	A	9.0	12.0
At elevated temperature	125°C	7.0	7.0
After exposure to processing solutions	A	7.0	12.0
2. Volume resistivity, minimum in megohm-centimeter	C-96/35/90	10^6	---
After moisture resistance	F	---	10^6
At elevated temperature	E-24/125	10^3	10^3
3. Surface resistivity, minimum megohm	C-96/35/90	10^4	---
After moisture resistance	F	---	10^4
At elevated temperature	E-24/125	10^3	10^3
4. Dimensional stability, maximum changes inches/inch		Class A	Class B
After stress	E4/105	0.00075	0.0005
After bake	E2/150	0.00075	0.0005
5. Water absorption, maximum percent	E-1/105+des +D-24/23	---	(inch) 0.031-0.80 / 0.062-0.35 / 0.093-0.25 / 0.125-0.20 / 0.250-0.13
6. Dielectric breakdown (parallel to laminations), minimum kV, step-by-step	D-48/50+ D-0.5/23	---	40
7. Electric strength, perpendicular to laminations, average, minimum volts/mil	D-48/50+ D-0.5/23	750	---
8. Permittivity at 1 MHz max	C-40/23/50	5.4	5.4

See footnote at end of table.

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TABLE I. Property values - Continued.

Property to be tested	Test condition 1/	Core thickness	
		Less than 0.020	0.020 in. and over
9. Loss tangent at 1 MHz max.	C-40-23/50	0.035	0.03
10. Q (resonance), minimum	D-24/23		
at 1 MHz		All	50
at 50 MHz		All	70
11. Flexural strength, minimum thousand lbs-per-square inch			
A. Lengthwise	A	---	60
B. Crosswise	A	---	50
C. Lengthwise at elevated temperature	N/A	N/A	N/A
12. Arc resistance, minimum seconds	D-48/50+ D-0.5/23	60	60
13. Flammability, maximum burntime, seconds, each flame application	N/A	N/A	N/A

1/ See specimen conditioning requirements, conditioning tolerances and testing information table XXVI of MIL-P-13949.

Marking Internal marking shall not be red in color (when specified).

NOTE: When approved by the acquiring activity, type GF may be used in lieu of type GE. (Type GF contains all of the properties of type GE with the addition of flame resistance).

NOTE: Revision letters are not used to denote changes due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - AS, OS, SH
Air Force - 11, 85
NSA - S5
DLA - ES

User activities:
Army - AT, ME
Navy - MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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